DESCRIPTIVE STATISTICS · STUDY NOTES

Standard Deviation and Coefficient of Variation

Measuring spread in the original units, and comparing spread across datasets

30-SECOND REVISION

Standard deviation = square root of the variance. It is in the **original units** (dollars, not dollars²), so it is directly interpretable.
Coefficient of variation = standard deviation ÷ mean. It has **no units**, so it lets you compare the spread of different datasets.
Pizza example: prices in dollars and pesos give different SDs (3.27 vs 61.59) but the **same CV ≈ 0.60**.

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1. Where this fits

Descriptive statistics summarise one variable with three families of measures. Variability has three common tools, each fixing a weakness of the one before it.

Family	Question	Measures
Central tendency	Where is the centre?	Mean, median, mode
Asymmetry	Is it lopsided?	Skewness
Variability	**How spread out is it?**	**Variance → standard deviation → coefficient of variation**

Measure	Fixes which problem?	Units
Variance	Measures spread, but...	Squared units (dollars²)
Standard deviation	...is hard to read, so take the square root	Original units (dollars)
Coefficient of variation	...cannot compare different datasets, so divide by the mean	None (a pure number)

These three, together with the measures of central tendency and asymmetry, are called **univariate measures** because each describes **one variable** at a time. (Covariance and correlation, covered later, describe two variables together.)

2. Standard deviation (SD)

Definition. The standard deviation is the square root of the variance. It tells you the typical distance of a data point from the mean.

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}}$$

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

	Population	Sample
Symbol	σ (sigma)	s
Built from	σ² (population variance)	s² (sample variance)
Denominator inside the root	N	n − 1

Why prefer SD over variance?

- **Variance is in squared units.** If prices are in dollars, variance is in dollars², which has no real-world meaning.
- **SD is in the original units.** A standard deviation of $3.27 means prices typically sit about $3.27 from the average price.
- That is why professionals use SD as the preferred measure of variability: it is **directly interpretable**.
- SD is the most common measure of variability **for a single dataset**.

> **The limitation of SD**
> SD depends on the units. The same prices in dollars and in pesos give **different** standard deviations, even though nothing about the real spread changed. So you cannot use SD to compare two datasets on different scales. That is the job of the coefficient of variation.

3. Coefficient of variation (CV)

Definition. The coefficient of variation divides the standard deviation by the mean. It is also called the **relative standard deviation**, because it expresses spread relative to the size of the average.

$$CV = \frac{s}{\bar{x}} \qquad CV = \frac{\sigma}{\mu}$$

Sample Population

- Dividing by the mean cancels the units, so the CV is **just a number** (for example 0.60), not dollars or pesos.
- Because it is unit-free, you can **compare the variability of two or more datasets**, which SD cannot do.

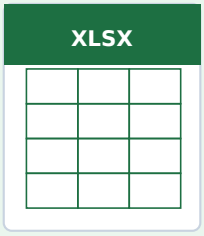
- A **higher** CV means more spread relative to the average. A **lower** CV means the data is more consistent.

> **Supplementary note (not in the class notes)**
> CV only makes sense when the mean is clearly above zero. If the mean is close to zero, the ratio becomes huge and unstable, and a mean of exactly zero makes it undefined. It is also meant for data measured on a ratio scale (a true zero), such as prices, heights or weights.

4. Worked example: pizza prices

The class example takes the price of a slice of pizza at **10 places in New York**. The prices range from **$1 to $11**. Then the same prices are converted into Mexican pesos, so there are **two datasets describing the same pizzas**.

4.1 The data set



click to open

02_Standard_Deviation.xlsx

DATA SET SENT
attached to this document

Excel workbook · 11,229 bytes (11.0 KB) · sheet “Std and cv”

10 pizza prices in dollars and pesos, plus mean, sample SD and CV

Attached inside this PDF. Click the file icon, or open your PDF viewer's Attachments panel. Keep a copy beside this PDF in the repo folder as well.

Place	1	2	3	4	5	6	7	8	9	10
Dollars ($)	1.00	2.00	3.00	3.00	5.00	6.00	7.00	8.00	9.00	11.00
Pesos (MXN)	18.81	37.62	56.43	56.43	94.05	112.86	131.67	150.48	169.29	206.91

Pesos = dollars × 18.81. In the workbook, column C holds the dollar prices (C5:C14) and column D computes the pesos with `=C5*18.81`.

4.2 The four steps

Step 1: sample or population?. The 10 places are not every pizzeria in New York, so this is a **sample**. Use the sample formulas (÷ n − 1).

Step 2: find the mean. Dollars: (1+2+3+3+5+6+7+8+9+11) ÷ 10 = 55 ÷ 10 = **$5.50**. Pesos: 5.50 × 18.81 = **MXN 103.46**.

Step 3: find the sample variance. Square each distance from the mean, add them up and divide by n − 1 = 9 (table below).

Step 4: take the square root. SD = √variance. This is the standard deviation, in the original units.

Squared distances from the mean (dollars)

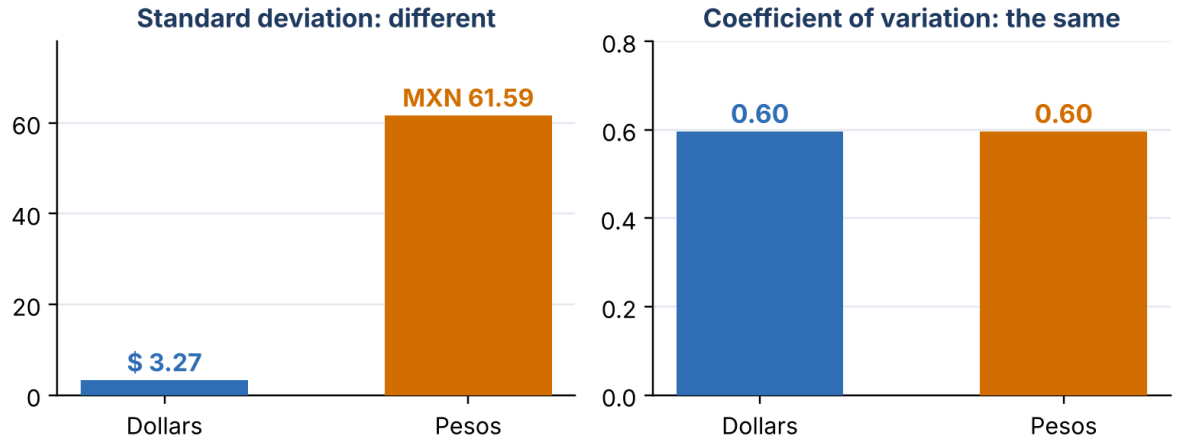
Price x ($)	x − 5.50	(x − 5.50)²
1.00	-4.50	20.25
2.00	-3.50	12.25
3.00	-2.50	6.25
3.00	-2.50	6.25
5.00	-0.50	0.25
6.00	+0.50	0.25
7.00	+1.50	2.25
8.00	+2.50	6.25
9.00	+3.50	12.25
11.00	+5.50	30.25
Sum	**0.00**	**96.50**

$$s^2 = \frac{96.5}{10-1} = 10.72 \qquad s = \sqrt{10.72} = 3.27 \qquad CV = \frac{3.27}{5.50} = 0.60$$

variance | standard deviation | coefficient of variation

4.3 Results side by side

Measure	Dollars ($)	Pesos (MXN)	Same?
Mean	$ 5.50	MXN 103.46	No (different units)
Sample variance	$² 10.72	MXN² 3793.69	No
Sample standard deviation	$ 3.27	MXN 61.59	No
Coefficient of variation	**0.60**	**0.60**	**Yes**



What this proves
Same pizzas, same restaurants, yet the standard deviation looks completely different ($3.27 vs 61.59 pesos). That seems wrong, because the real spread has not changed. Dividing by each dataset's own mean gives **0.60 for both**, so we correctly conclude that the two datasets have **the same variability**.

Why does the CV stay the same?

Converting dollars to pesos multiplies every price by the same number, 18.81. The mean and the SD are both multiplied by 18.81, so the 18.81 cancels in SD ÷ mean. The variance, however, is multiplied by $18.81^2 \approx 353.82$ ($10.72 \times 353.82 \approx 3793.69$). This is a handy check: **if you rescale data by a factor c, then mean and SD scale by c, variance scales by c^2, and CV does not change.**

Reading the SD on the data



The shaded band runs one standard deviation either side of the mean (2.23 to 8.77). **6 of the 10 prices** fall inside it. The band is a useful picture of a "typical" price range. *This chart was added for illustration; it is not from the class.*

5. Excel and Python

Goal	Excel	Python (statistics)	NumPy
Mean	=AVERAGE(C5:C14)	mean(data)	np.mean(data)
Sample SD (÷ n − 1)	=STDEV.S(C5:C14)	stdev(data)	np.std(data, ddof=1)
Population SD (÷ N)	=STDEV.P(range)	pstdev(data)	np.std(data)
Sample variance	=VAR.S(range)	variance(data)	np.var(data, ddof=1)
Sample CV	=G7/G5	stdev(d)/mean(d)	np.std(d, ddof=1)/np.mean(d)

```
import statistics as st

usd = [1, 2, 3, 3, 5, 6, 7, 8, 9, 11]
mxn = [x * 18.81 for x in usd]

for name, d in [("dollars", usd), ("pesos", mxn)]:
    mean, sd = st.mean(d), st.stdev(d)       # stdev = sample SD (n - 1)
    print(name, round(mean, 2), round(sd, 2), round(sd / mean, 2))

# dollars 5.5     3.27   0.6
# pesos   103.45  61.59  0.6      <- the mean is exactly 103.455
```

Rounding note: the pesos mean is 103.455. Python prints 103.45 (binary floating point) while Excel displays 103.46. Both are the same value, rounded differently.

> **Pitfall**
> `np.std` divides by N unless you pass `ddof=1`. Python's `statistics.stdev` and pandas' `Series.std()` already divide by n − 1. In Excel, **STDEV.S** is the sample version and **STDEV.P** the population version.

6. Revision kit

Formula sheet

Quantity	Sample	Population
Mean	$\bar{x} = \Sigma x \div n$	$\mu = \Sigma x \div N$
Variance	$s^2 = \Sigma(x - \bar{x})^2 \div (n - 1)$	$\sigma^2 = \Sigma(x - \mu)^2 \div N$
Standard deviation	$s = \sqrt{s^2}$	$\sigma = \sqrt{\sigma^2}$
Coefficient of variation	$CV = s \div \bar{x}$	$CV = \sigma \div \mu$

Self-test (cover the answers and try first)

#	Question	Answer
1	Why is SD preferred to variance?	SD is in the original units, so it is directly interpretable. Variance is in squared units.
2	Why can't SD compare two datasets in different units?	SD changes when the units change (dollars vs pesos), even though the real spread is identical.
3	What does the CV remove, and how?	The units, by dividing the SD by the mean.
4	If every value is multiplied by 10, what happens to the mean, SD, variance and CV?	Mean × 10, SD × 10, variance × 100, CV unchanged.
5	Data: 2, 4, 6, 8 (a sample). Find s and CV.	Mean 5. $\Sigma(x - 5)^2 = 20$, $s^2 = 20 \div 3 = 6.67$, $s = 2.58$, $CV = 0.52$.
6	Dataset A: mean 50, SD 5. Dataset B: mean 10, SD 2. Which varies more relative to its mean?	A: CV = 0.10. B: CV = 0.20. **B** is more variable, even though its SD is smaller.

Common mistakes

- Comparing the **standard deviations** of datasets in different units instead of their CVs.
- Dividing by N when the data is a **sample** (use n − 1).
- Reading variance as if it were in the original units. It is in **squared** units.
- Quoting the CV with units. It has **none**.
- Using the CV when the mean is near zero.

7. Notes check: points to double-check in the class notes

While preparing these notes I compared the class notes with the workbook. These small inconsistencies are worth correcting in your own copy; the numbers in this document follow the workbook.

Class notes say	The workbook shows	Suggested fix
“only 11 restaurants in NY” and also “n = 10”	10 prices (C5:C14), ranging from $1 to $11	The sample has **n = 10**. The 11 is the highest price, not a count.
“exchange rate from 1 dollar 18.81 to 206.91”	Pesos = dollars × 18.81. So $1 = 18.81 pesos and $11 = 206.91 pesos.	The exchange rate is **18.81 pesos per dollar**. The 18.81 to 206.91 is the **price range** in pesos.
Variance is “calculated”	Cells G6 and H6 hold the variance as **typed text** (“$2 10.72”, “MXN2 3793.69”), not a formula	Replace them with `=VAR.S(C5:C14)` and `=VAR.S(D5:D14)` so they update with the data.

In one line each

- **Standard deviation** = √variance, in the original units; use it to describe one dataset.
- **Coefficient of variation** = SD ÷ mean, unit-free; use it to compare datasets.
- Pizza example: SDs differ (3.27 vs 61.59), CVs match (0.60), so the variability is the same.

Source: class notes (Part 4) and the Excel file *02_Standard_Deviation.xlsx*, which is attached on page 3 (section 4.1). Sections 4.3 (the scaling explanation), the supplementary note on CV, the “Reading the SD” chart, the revision kit and the notes check were added for clarity. All numbers were checked by computation against the workbook.